

Data Dictionary

Project Name

Pharma Sales Force Effectiveness Analytics System

Purpose

This document defines all data fields used in the Pharma Sales Force Effectiveness Analytics System. The dataset is designed to analyze sales performance, territory effectiveness, physician contribution, product performance, and marketing ROI.

Column Name	Data Type	Description	Business Purpose
Sales_ID	Integer (PRIMARY KEY)	Unique identifier for each sales transaction	Used to uniquely track sales records and avoid duplicates
Date	Date	Transaction date of the sales activity	Used for trend and time-series analysis
Region	Text	Sales territory or geographic region	Used for territory performance analysis
Sales_Rep_Name	Text	Name of the sales representative	Used to evaluate sales force effectiveness
Doctor_ID	Integer	Unique identifier for each physician	Used to uniquely identify doctors and avoid duplicate records
Doctor_Name	Text	Name of the physician	Used for physician analysis and segmentation
Specialty	Text	Medical specialty of the physician	Used for specialty-wise analysis
Medicine_Name	Text	Name of the pharmaceutical product	Used for product performance analysis
Units_Sold	Integer	Number of medicine units sold	Used to calculate revenue and product demand

Column Name	Data Type	Description	Business Purpose
Revenue	Decimal	Revenue generated from product sales	Primary business KPI
Prescriptions	Integer	Number of prescriptions generated	Used to identify high-value physicians
Marketing_Spend	Decimal	Marketing expenditure allocated to campaigns or territories	Used for ROI analysis
Target_Sales	Decimal	Sales target assigned to a sales representative	Used to evaluate performance against goals
Achieved_Pct	Decimal	Percentage of target achieved by a sales representative	Used for productivity and ranking analysis
ROI	Decimal	Return generated for every unit of marketing spend	Used to measure marketing effectiveness

Derived Metrics

Revenue

Definition:

Total sales generated from pharmaceutical products.

Formula:

Revenue = Units Sold × Unit Price

Achievement Percentage

Definition:

Measures how effectively a sales representative achieves assigned targets.

Formula:

Achievement % = (Actual Sales ÷ Target Sales) × 100

Prescription Volume

Definition:

Total prescriptions generated by physicians.

Formula:

Prescription Volume = Sum(Prescriptions)

Marketing ROI

Definition:

Measures revenue generated for every unit of marketing spend.

Formula:

ROI = Revenue ÷ Marketing Spend

Revenue by Territory

Definition:

Total revenue generated by a specific territory or region.

Formula:

Revenue by Territory = Sum(Revenue)

Rep Productivity

Definition:

Measures sales performance of individual sales representatives.

Formula:

Rep Productivity = Revenue Generated by Representative

Data Quality Rules

1. Revenue must be greater than or equal to zero.
2. Marketing Spend must be greater than or equal to zero.
3. Prescription count cannot be negative.
4. Target Sales must be greater than zero.
5. Achievement Percentage cannot be negative.

6. Region values must not be null.
7. Sales Representative names must not be null.
8. Doctor names must not be null.
9. Product names must not be null.
10. Transaction Date must be valid and available for every record.

Dataset Usage

The dataset supports analysis in the following business areas:

- Territory Performance Analysis
- Sales Force Effectiveness Analysis
- Physician Segmentation Analysis
- Product Performance Analysis
- Marketing ROI Analysis
- Executive KPI Reporting
- Interactive Business Intelligence Dashboards